

Deliverability & Carrier Best Practices

Operational standards for reliable U.S. SMS/MMS delivery on RelayFox.

VERSION	2.2	UPDATED	March 12, 2026
PRIMARY ROUTE	Verified 10DLC	OWNER	Nina Alvarez, Messaging Operations
BRAND EXAMPLE	Juniper & Coast	PRODUCTION TPS	75 TPS approved

Corpus notice: This is a synthetic demonstration document created for Cortex Search evaluation. Brand names, people, metrics, URLs, short codes, and incidents are fictional.

1. 10DLC registration and route hygiene

Each U.S. long-code program must map to a verified brand and a registered campaign use case before production traffic begins. For Juniper & Coast, RelayFox uses brand profile JC-US-88421 and marketing campaign JC-MKT-2025-4470. The registered sample messages include welcome, cart, back-in-stock, and promotional broadcast content.

- Keep legal business name, EIN, website, privacy policy, terms, and opt-in flow consistent across registration records.
- Do not send affiliate offers, unrelated lead-generation messages, or high-risk vertical content under a standard marketing campaign.
- Re-register when message purpose materially changes. Copy testing within the same disclosed use case does not require a new campaign.
- Use production routes only. Sandbox or fallback routes must be blocked from audiences above 2,000 recipients.

2. Throughput planning

Audience	Approved TPS	Expected send time	Operational action
<= 5,000	25 TPS	<= 3.5 minutes	Standard queue; verify delivery sample after 500 sends
5,001-50,000	75 TPS	2-12 minutes	Run route/TPS preflight and projected quiet-hour check
50,001-250,000	150 TPS aggregate	6-28 minutes	Shard across approved routes; preserve per-carrier pacing
MMS above 20,000	75 TPS plus media validation	5+ minutes	Pre-cache asset and verify image fetch from carrier test devices

Estimated completion time is recipient count divided by effective TPS, adjusted for carrier pacing and retry reserves. A 23,804-recipient MMS batch at 75 TPS should clear in roughly 5-7 minutes. During RF-INC-2025-0822-PNW, a 3 TPS fallback route extended completion to 134 minutes and reduced campaign response.

3. Deliverability thresholds

Metric	Healthy	Watch	Stop / investigate
Delivery rate	>= 97.0%	95.0%-96.9%	< 95.0%
Carrier filtering	< 0.6%	0.6%-1.2%	> 1.2%
Hard invalid rate	< 1.0%	1.0%-2.0%	> 2.0%
Campaign opt-out rate	< 0.8%	0.8%-1.2%	> 1.2%
Complaint / abuse signals	< 0.02%	0.02%-0.05%	> 0.05%

4. Spam avoidance and content design

- Identify the sender early. Put "Juniper & Coast" in the first sentence or MMS subject line.
- Match expectation. Message purpose and frequency should resemble the opt-in disclosure and registered 10DLC use case.
- Avoid pressure patterns. Do not combine excessive capitalization, repeated punctuation, misleading countdowns, or unsupported "guaranteed" claims.
- Use clean personalization. Default missing first names to "friend" or omit the name. Never expose raw merge syntax to recipients.
- Control frequency. Rising opt-outs and reduced clicks are stronger warning signals than raw send volume alone.
- Warm new routes. Begin at 10,000-20,000 messages per day and increase only when delivery, filtering, and opt-out metrics remain healthy.

5. Link and domain policy

All promotional links must use a dedicated, HTTPS-enabled branded domain controlled by the sending brand or RelayFox. Juniper & Coast uses `rly.jnpr.example`. Public shorteners such as generic shared domains are prohibited because they create reputation coupling and obscure destination ownership.

- The first redirect must resolve in under 800 ms at the 95th percentile and use no more than two total redirects.
- Destination domain, certificate, and page content must be live before the campaign enters final approval.
- Do not rotate link domains to bypass filtering. Reputation issues should be corrected through consent, content, and traffic-quality changes.
- Append only approved analytics parameters. Never place phone numbers, email addresses, or raw customer IDs in query strings.

6. MMS media policy

Recommended creative is JPEG or PNG, 640 px wide, under 500 KB, with a simple focal point and readable safe-area text. RelayFox pre-fetches and caches media before launch. Animated GIFs and files above 800 KB require a carrier test because transcoding can alter quality or delay fetches.

7. Launch runbook

- 1 Confirm registered brand/campaign mapping and production route.
- 2 Calculate projected completion time at approved TPS.
- 3 Send seed messages to AT&T, T-Mobile, and Verizon test devices.
- 4 Validate sender name, merge tags, STOP footer, link redirect, and landing-page offer.
- 5 Watch delivery and filtering after the first 1,000 messages and at 25%, 50%, and 100% completion.
- 6 Pause on stop thresholds; do not switch domains or routes without Messaging Operations approval.